The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 JAN 94.

INCH-POUND

MIL-S-19500/498B
25 October 1993
SUPERSEDING
MIL-S-19500/498A
19 June 1990

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPE 2N6306, 2N6308, JANTX AND JANTXV

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3. (Similar to TO-3)

1.3 Maximum ratings.

	P_T T_C = +25°C 1/	P_T T_C = +100°C 1/	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}
	W	W	V dc	V dc	V dc	A dc	A dc	°C
2N6306	125	62.5	500	250	8.0	4.0	8.0	-65 to +200
2N6308	125	62.5	700	350	8.0	4.0	8.0	-65 to +200

1/ Between T_C = +25°C and T_C = +175°C, linear derating factor (average) = .833 W/°C.

1.4 Primary electrical characteristics.

	h_{FE2}		h_{FE3}		$V_{BE(sat)}$ 1/		$V_{CE(sat)}$ 1/			
	V_{CE} = 5 V dc, I_C = 3 A dc		V_{CE} = 5 V dc, I_C = 8 A dc		I_C = 8 A dc, I_B = 2/		I_C = 8 A dc, I_B = 2/		I_C = 3 A dc, I_B = 0.6 A dc	
	Min	Max	Min	Max	Min	Max	Min V dc	Max V dc	Min V dc	Max V dc
2N6306	15	75	4			2.3		5		0.8
2N6308	12	60	3			2.5		5		1.5

	$\|h_{fe}\|$	C_{obo}	Switching	
	V_{CE} = 10 V dc, I_C = 0.3 A dc, f = 1 MHz	V_{CB} = 10 V dc, I_E = 0, 100 ≤ f ≤ 1 MHz	t_{on}	t_{off}
		pF	µs	µs
Min	5			
Max	30	250	0.6	3.0

1/ Pulsed (see 4.5.1).
2/ 2N6306 (I_B) = 2.0 A dc; 2N6308 (I_B) = 2.67 A dc.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : NASA/Parts Project Office (NPPO), ATTN: NASA Goddard Space Flight Center, Code 311.A, Greenbelt, MD 20771 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 11.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening. Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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Screen (see table II of MIL-S-19500)	Measurements JANTX, JANTXV levels
1/	Thermal response (see 4.3.2)
11	I_{CEX1} and h_{FE3}
12	See 4.3.1
13	Subgroup 2 of table I herein. $\Delta I_{CEX1} \leq$ 100% of initial value or 500 nA dc, whichever is greater; $\Delta h_{FE3} \leq$ 25% of initial value.

1/ This test shall be performed any time before screen 10.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; $V_{CB} \geq$ 10 V dc; T_J = +162.5°C ±12.5°C. No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal response ΔV_{BE} measurements. The ΔV_{BE} measurements shall be performed in accordance with MIL-STD-750, method 3131. The ΔV_{BE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor. The chosen ΔVBE measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. One hundred percent Safe Operation Area (SOA) testing may be performed in lieu of thermal response testing herein, provided that the appropriate conditions of temperature, time, current, and voltage to achieve die attach integrity are submitted to the qualifying activity. The following parameter measurements shall apply:

a. I_H measurement - - - - - - - - - - - - - - - - - 10 mA.

b. V_{CE} measurement voltage - - - - - - - - - - - - 20 V (same as V_H).

c. I_H collector heating current - - - - - - - - - - 4 A (minimum).

d. V_H collector-emitter heating voltage - - - - - - 20 V (minimum).

e. t_H heating time - - - - - - - - - - - - - - - - - 100 ms.

f. t_{MD} measurement delay time - - - - - - - - - - - 50 µs to 80 µs.

g. t_{SW} sample window time - - - - - - - - - - - - - 10 µs (maximum).

4.3.3 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. I_H measurement - - - - - - - - - - - - - - - - - 10 mA.

b. V_{CE} measurement voltage - - - - - - - - - - - - 15 V.

c. I_H collector heating current - - - - - - - - - - 8 A.

d. V_H collector-emitter heating voltage - - - - - - 15 V.

e. t_H heating time - - - - - - - - - - - - - - - - - Stead-state (see MIL-STD-750, method 2131 for definition).

f. t_{MD} measurement delay time - - - - - - - - - - - 20 µs (maximum).

g. t_{SW} sample window time - - - - - - - - - - - - - 10 µs (maximum).

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
3	1037	$V_{CB} \geq 10$ V dc; ΔT_J between cycles $\geq +100°C$; $t_{on} = t_{off} \geq 1$ minute for 2000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.3.1. Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
6	1037	$V_{CB} \geq 10$ V dc; ΔT_J between cycles $\geq +100°C$; $t_{on} = t_{off} \geq 1$ minute for 6000 cycles. No heat sink or forced-air cooling on device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to base breakdown voltage	3011	Bias condition D; I_C = 100 mA dc; Pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6306				250		V dc
2N6308				350		V dc
Collector to emitter cutoff current	3041	Bias condition D;	I_{CEO}		50	µA dc
2N6306		V_{CE} = 250 V dc				
2N6308		V_{CE} = 350 V dc				
Emitter-base cutoff current	3061	Bias condition D; V_{EB} = 8 V dc	I_{EBO}		5.0	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc;	I_{CEX1}			
2N6306		V_{CE} = 500 V dc			5.0	µA dc
2N6308		V_{CE} = 700 V dc			5.0	µA dc
Base emitter voltage	3066	Test condition A; I_C = 8.0 A dc; Pulsed (see 4.5.1)	$V_{BE(sat)}$			
2N6306		I_B = 2.0 A dc			2.3	V dc
2N6308		I_B = 2.67 A dc			2.5	V dc
Base emitter voltage	3066	Test condition B; I_C = 3.0 A dc; V_{CE} = 5.0 V dc; Pulsed (see 4.5.1)	$V_{BE(on)2}$			
2N6306					1.3	V dc
2N6308					1.5	V dc
Collector to emitter saturated voltage	3071	I_C = 8.0 A dc; Pulsed (see 4.5.1)	$V_{CE(sat)1}$		5.0	V dc
2N6306		I_B = 2.0 A dc				
2N6308		I_B = 2.67 A dc				
Collector to emitter saturated voltage	3071	I_C = 3.0 A dc; I_B = 0.6 A dc; Pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N6306					0.8	V dc
2N6308					1.5	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 3.0 A dc; Pulsed (see 4.5.1)	h_{FE1}			
2N6306				15	75	
2N6308				12	60	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 8.0 A dc; Pulsed (see 4.5.1)	h_{FE2}			
2N6306				4		
2N6308				3		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 0.5 A dc; Pulsed (see 4.5.1)	h_{FE3}			
2N6306				15		
2N6308				12		
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX2}		500	µA dc
2N6306		V_{CE} = 450 V dc				
2N6308		V_{CE} = 650 V dc				
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 3.0 A dc; Pulsed (see 4.5.1)	h_{FE4}			
2N6306				6		
2N6308				5		
Subgroup 4						
Pulse response: transfer ratio	3251	Test condition A except test circuit and pulse requirements in accordance with figure 1.				
Turn-on time		V_{CC} = 125 V dc; I_C = 3.0 A dc; I_B = 0.6 A dc	t_{on}		0.6	µs

See footnote at end of table.

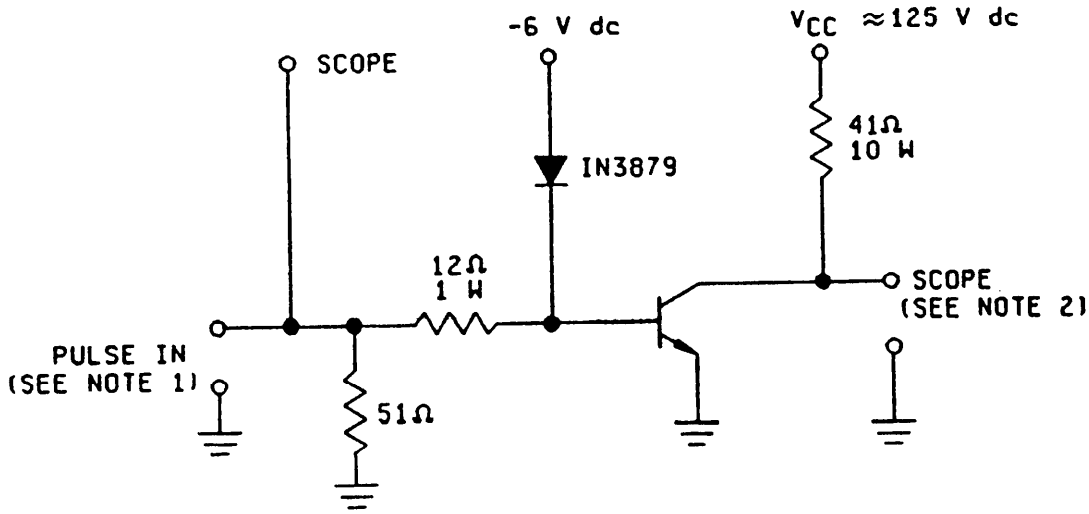
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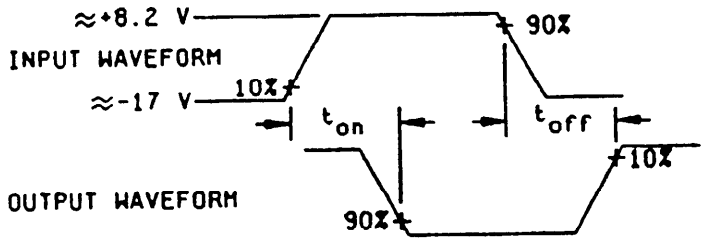
TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Turn-off time		V_{CC} = 125 V dc; I_C = 3.0 A dc; I_{B1} = 0.6 A dc I_{B2} = 1.5 A dc	t_{off}		3.0	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 0.3 A dc; f = 1 MHz	$\|h_{fe}\|$	5	30	
Open capacitance (open circuit)	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{obo}		250	pF
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 4.0 V dc; I_C = 0.5 A dc; f = 1.0 kHz	h_{fe}	5		
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C; t = 1 s; 1 cycle; (See figures 2 and 3)				
Test 1 (Both device types)		V_{CE} = 15.6 V dc; I_C = 8 A dc				
Test 2 (Both device types)		V_{CE} = 37 V dc; I_C = 3.4 A dc				
Test 3 2N6306		V_{CE} = 200 V dc; I_C = 65 mA dc				
2N6308		V_{CE} = 300 V dc; I_C = 25 mA dc				
Electrical measurements		See table II, steps 1 and 3				
Subgroups 6 and 7						
Not applicable						

For sampling plan see MIL-S 19500.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2 ns; duty cycle ≤ 1 percent; generator source impedance shall be 50Ω; pulse width = 30 μs.
2. Output sampling oscilloscope: Z_{IN} ≥ 20 kΩ; C_{IN} ≤ 50 pF; rise time ≤ 0.2 ns.

FIGURE 1. Pulse response test circuit.

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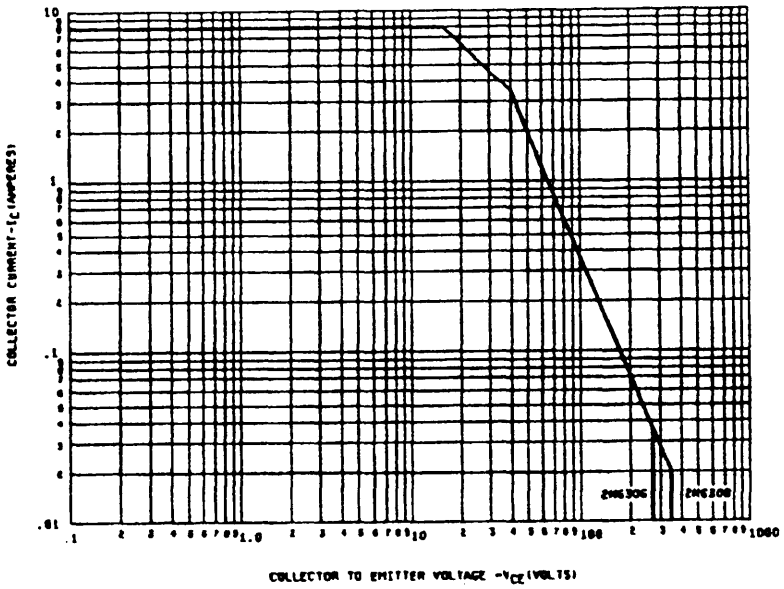


FIGURE 2. Maximum safe operating area graph (continuous dc).

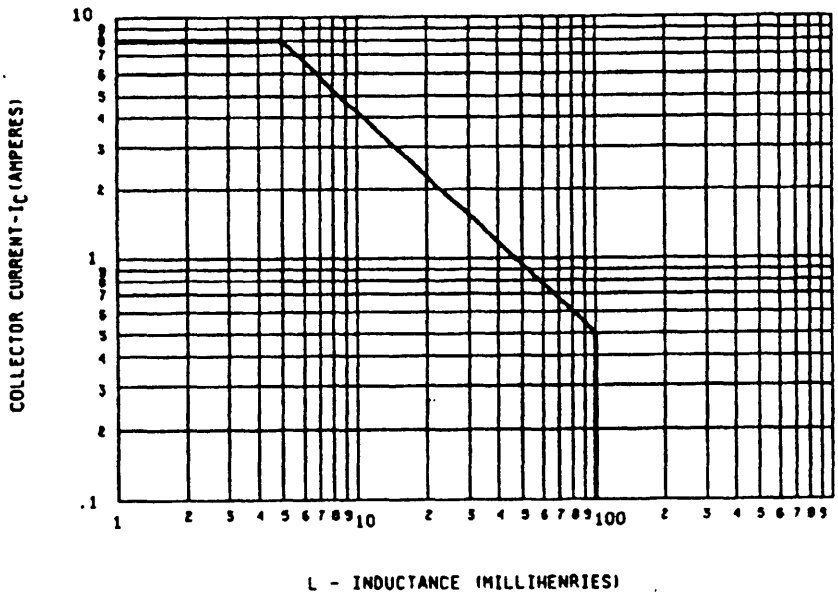


FIGURE 3. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 17
NASA - NA

Review activities:
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA-ES

(Project 5961-1499)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/4988	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER, TYPES 2N6306, 2N6308, JANTX AND JANTXV.

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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